

સંચાર ઈજનેરી (Communication Engineering) — સંપૂર્ણ સ્ટડી નોટ્સ

GPSC | GSSSB | PSI | Wireless PSI | Class-2 & Class-3 Exam માટે

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1 Introduction to Communication Systems

સંચાર (Communication) એટલે Source થી Destination સુધી માહિતી (Voice, Data, Image, Video) નું પ્રસારણ.

૩ મૂળભૂત ઘટકો: Transmitter (પ્રેષક) → Channel (માધ્યમ) → Receiver (પ્રાપ્તકર્તા)

Block Diagram: Information Source → Transducer → Transmitter → Channel (Noise) → Receiver → Transducer → Destination

પ્રકારો

આધાર	પ્રકારો
દિશા	Simplex (એક દિશા) / Half Duplex (વારંફરતી બંને) / Full Duplex (એકસાથે બંને)
Signal	Analog / Digital
માધ્યમ	Wired (Guided) / Wireless (Unguided)

Analog vs Digital

બાબત	Analog	Digital
Signal	Continuous	Discrete (0,1)
Noise Immunity	ઓછી	વધુ (Regeneration શક્ય)
Bandwidth	ઓછી	વધુ
Security	ઓછી	વધુ (Encryption)

Key Terms

- Bandwidth:** Highest-Lowest Frequency નો તફાવત
- Noise:** અનિચ્છનીય Signal
- Attenuation:** Signal Strength ઘટવી
- Modulation:** Message ને Carrier પર આરોપિત કરવું
- SNR:** Signal Power / Noise Power

📌 **Modulation શા માટે?** (1) Antenna Size ઘટાડવા (2) Multiplexing શક્ય બનાવવા (3) Noise ઘટાડવા (4) Range વધારવા

2 Modulation — AM, FM, PM

Carrier: $c(t) = A_c \sin(2\pi f_c t + \phi)$ — Amplitude(A_c), Frequency(f_c), Phase(ϕ) — આ ૩ માંથી બદલવાથી ૩ પ્રકાર બને છે.

Parameter બદલાય	Modulation
Amplitude	AM
Frequency	FM
Phase	PM

AM (Amplitude Modulation)

- Modulation Index: $m_a = A_m/A_c$ (૦ થી ૧, $m_a > 1 \rightarrow$ Over Modulation)
- BW = 2f_m**
- ૩ Components: Carrier(f_c) + USB($f_c + f_m$) + LSB($f_c - f_m$) — માહિતી ફક્ત Side Bands માં, Carrier Power વેડફાય
- પ્રકારો: DSB-FC (સામાન્ય), DSB-SC, SSB, VSB
- ઉપયોગ:** AM Radio (535-1605 kHz), Aircraft Communication

FM (Frequency Modulation)

- Frequency Deviation: $\Delta f = k_f \times A_m$
- Modulation Index: $\beta = \Delta f / f_m$
- Carson's Rule: **BW = 2(Δf + f_m) = 2f_m(β+1)**
- NBFM: $\beta \leq 0.5$, WBFM: $\beta > 1$
- ફાયદો:** Noise Immunity ઉત્તમ (Limiter વાપરી Amplitude Noise Remove), Capture Effect
- ઉપયોગ:** FM Radio (88-108 MHz), TV Sound, Mobile

PM (Phase Modulation)

- FM = PM of Integral of message; PM = FM of Derivative of message
- ઉપયોગ:** PSK, Digital/Satellite Communication

AM vs FM vs PM

લક્ષણ	AM	FM	PM
બદલાવું	Amplitude	Frequency	Phase
BW	ઓછી (2f _m)	વધુ (Carson)	વધુ

લક્ષણ	AM	FM	PM
Noise Immunity	ઓછી	વધુ સારી	વધુ સારી
Complexity	સરળ	જટિલ	જટિલ

3 AM Receiver & FM Receiver

Receiver ની 4 Performance Parameters: Sensitivity, Selectivity, Fidelity, Image Frequency Rejection

Superheterodyne Receiver (Block Diagram)

Antenna → RF Amp → Mixer → IF Amp → Detector → AF Amp → Speaker (↑ Local Oscillator)

Block	કાર્ય
RF Amplifier	નબળા Signal ને Amplify + Preliminary Selectivity
Mixer + LO	RF ને IF માં Convert (Heterodyning)
IF Amplifier	Fixed Frequency પર મુખ્ય Gain + Selectivity
Detector	Original Signal પાછો મેળવે
AGC	Output Volume Constant રાખે

AM Standard IF = 455 kHz | FM Standard IF = 10.7 MHz

Image Frequency ★

$f_{\text{image}} = f_{\text{RF}} + 2 \cdot f_{\text{IF}}$ — ઊંચી IF → સારો Image Rejection પણ Selectivity ઘટે (Trade-off)

AM vs FM Receiver

બાબત	AM Receiver	FM Receiver
IF	455 kHz	10.7 MHz
Bandwidth	સાંકડી	પહોળી
Detector	Envelope Detector	Discriminator/Ratio Detector
વધારાનું Block	AGC	Limiter + De-emphasis (75 μs)

📌 FM માં **Limiter** Amplitude Noise Remove કરે → Noise Immunity ઉત્તમ

4 Antenna & Wave Propagation

Antenna = Transducer (Electrical ↔ EM Wave), Reciprocal Device

Antenna Parameters

Radiation Pattern, Gain(G), Directivity(D), Efficiency($\eta=G/D$), Beamwidth, Impedance, Polarization

Antenna પ્રકારો

Half-wave Dipole ($\lambda/2$), Quarter-wave Monopole ($\lambda/4$), Yagi-Uda (Directional), Parabolic Dish (High Gain), Horn (Microwave)

Wave Propagation ની 3 પ્રકાર

પ્રકાર	Frequency	માધ્યમ	Range	ઉપયોગ
Ground Wave	LF/MF (30kHz-3MHz)	Earth Surface Follow	મર્યાદિત	AM Local Broadcasting
Sky Wave	HF (3-30MHz)	Ionosphere Reflection	ખૂબ લાંબુ	Short Wave Radio
Space Wave	VHF+ (>30MHz)	Line of Sight (LOS)	Antenna Height આધારિત	TV, FM, Satellite, Mobile

Ionosphere Layers: D, E, F1, F2 (F2 સૌથી અગત્યનું — રાત્રે પણ હાજર)

✦ D-Layer રાત્રે અદ્રશ્ય → રાત્રે AM Long Distance Reception સારું

Critical Frequency: $f_c = 9\sqrt{N_{max}}$ | MUF = $f_c \times \sec \theta$

LOS Distance: $d(km) = 4.12(\sqrt{h_t} + \sqrt{h_r})$ — Antenna Height વધુ → Range વધુ

5 SSB Techniques & SSB Receiver

SSB = DSB-SC + એક Side Band Filter Out (Carrier + એક Side Band Suppress)

ફાયદા

BW = fm (અસધું), Power Saving $\approx 83\%$, Better SNR, ઓછો Fading

Generation Methods

1. Filter Method — Balanced Modulator → Sharp Band Pass Filter (Crystal/Ceramic Filter)

2. **Phase Shift Method** — 90° Phase Shift Network + 2 Balanced Modulator, Filter જરૂરી નથી
3. **Third Method (Weaver's)** — Filter+Phasing જુગ Combination

SSB Receiver

Detection = **Product Detector** (Envelope Detector નહીં) + **BFO (Beat Frequency Oscillator)** જરૂરી — Local Carrier Regenerate કરેલ

🚩 BFO Frequency Error → "Donald Duck" Audio Distortion

લક્ષણ	AM Receiver	SSB Receiver
Detector	Envelope	Product Detector
Local Carrier	જરૂરી નથી	જરૂરી (BFO)
IF BW	પહોળી	સીકડી (~3kHz)

ઉપયોગ: HF Communication, Amateur Radio, Marine, FDM Telephone

6 Communication Receiver

Broadcast Receiver કરતાં વધુ Sensitivity, Selectivity, Stability — HF/Amateur/Military/Marine માટે

Double Conversion Superheterodyne: 1st IF (ઊંચી, Image Rejection માટે) + 2nd IF (નીચી, Selectivity માટે — Crystal Filter)

વિશિષ્ટ Features

Feature	કાર્ય
S-Meter	Signal Strength દર્શાવે
BFO	SSB/CW Detection
Squelch	Signal ન હોય ત્યારે Speaker Mute
Noise Limiter (ANL)	Impulse Noise Remove
Frequency Synthesizer (PLL)	Crystal Reference સાથે Stable Frequency

7 Line Coding Schemes

વ્યાખ્યા: Binary Data ને Transmission માટે Electrical Signal માં Convert કરવું

સારી Scheme ના લક્ષણો

No DC Component, Self-Synchronization, Error Detection, Noise Immunity, Bandwidth Efficiency

પ્રકારો

Scheme	વર્ણન	DC	Sync	વપરાશ
Unipolar NRZ	'1' = +V, '0' = 0V	હાજર	Poor	જૂની Systems
Polar NRZ	'1' = +V, '0' = -V	ઓછો	Poor	—
Polar RZ	Mid-bit 0 પર પાછું	નહી	Good	—
Manchester	Mid-bit Transition	None	Excellent	Ethernet
Diff. Manchester	Bit-start Transition	None	Excellent	Token Ring
AMI (Bipolar)	'1' Alternately +V/-V	None	Moderate	Telephone Lines
HDB3	AMI + Zero Substitution	None	Good	PCM (E1 Lines — India/Europe)

🔴 B8ZS (USA) vs HDB3 (Europe/India) — બંને AMI ની Long-Zero Sync Problem Solve કરે

8 Pulse Modulation & Digital Modulation

Sampling Theorem (Nyquist)

$f_s \geq 2f_m$ — $f_s = 2f_m$ ને Nyquist Rate કહેવાય; $f_s < 2f_m \rightarrow$ Aliasing 🔴 Voice ($f_m = 4\text{kHz}$) $\rightarrow$ Standard Sampling Rate = 8 kHz

Analog Pulse Modulation

પ્રકાર	બદલાતું Parameter	Noise Immunity
PAM	Amplitude	ઓછી
PWM/PDM	Width	સારી
PPM	Position	સૌથી સારી

PCM (Pulse Code Modulation) – Digital

3 પગલાં: Sampling → Quantization → Encoding

- Quantization Levels: $L = 2^n$
- SQNR: $SQNR = 6.02n + 1.76 \text{ dB}$ (દરેક Bit → 6dB સુધારો)
- Bit Rate = $f_s \times n \rightarrow$ Telephone: $8000 \times 8 = 64 \text{ kbps}$
- Companding: μ -law (USA/Japan), A-law (Europe/India)

Digital Modulation (ASK, FSK, PSK)

પ્રકાર	બદલાતું	Noise Immunity	BW
ASK (OOK)	Amplitude	સૌથી ઓછી	2Rb
FSK	Frequency	મધ્યમ	$2(\Delta f + R_b)$
PSK	Phase	સૌથી વધુ	Coherent Detection જરૂરી

✦ QPSK = 2 bits/symbol → BPSK કરતાં બમણી Bandwidth Efficiency ✦ AM↔ASK, FM↔FSK, PM↔PSK (Analog-Digital Correspondence)

9 Source Coding & Multiplexing

Source Coding

Entropy: $H = \sum p_i \log_2(1/p_i)$ bits/symbol (Theoretical Minimum Bits) Coding

Efficiency: $\eta = (H/L_{avg}) \times 100\%$

Technique	સિદ્ધાંત	વપરાશ
Shannon-Fano	Top-Down Divide (Probability Sort)	—
Huffman	Bottom-Up Combine — સૌથી Optimal	ZIP, JPEG, MP3
RLE	પુનરાવર્તિત Symbol → Count	Fax

Multiplexing (MUX/DEMUX)

પ્રકાર	આધાર	Guard	વપરાશ
FDM	Frequency	Guard Band	Radio, Cable TV
TDM	Time	Guard Time	Digital Telephone

પ્રકાર	આધાર	Guard	વપરાશ
WDM	Wavelength (Light)	—	Optical Fiber
CDM	Unique Code	—	CDMA Mobile, GPS

✦ FDM = Analog માટે યોગ્ય, TDM = Digital માટે યોગ્ય

10 Data Communication

5 ઘટકો: Message, Sender, Receiver, Medium, Protocol

Bit Rate vs Baud Rate: $\text{Bit Rate} = \text{Baud Rate} \times \log_2(L)$ (L = Signal Levels)

Codes

ASCII (7-bit, 128 char), EBCDIC (8-bit, IBM), Unicode (16-bit+)

Transmission Mode

પ્રકાર	વર્ણન
Parallel	એકસાથે Multiple Bits, ટૂંકા અંતર, ઝડપી
Serial	એક પછી એક Bit, લાંબા અંતર
Asynchronous	Start/Stop Bit સહિત Character-wise
Synchronous	Clock-based Block Transfer, Overhead ઓછો

Transmission Impairments

Attenuation ($\text{dB} = 10 \log_{10}(P_1/P_2)$), Distortion, Noise

Transmission Media

Twisted Pair < Coaxial < Optical Fiber (Bandwidth ક્રમમાં)

OSI Model (7 Layers) ★

Physical → Data Link → Network → Transport → Session → Presentation → Application

Trick: "Please Do Not Throw Sausage Pizza Away"

1 1 Error Detection Techniques

Technique	સિદ્ધાંત	Detects
Parity Check	'1' ની સંખ્યા Even/Odd	Single-bit only
2D Parity	Row+Column Parity	Single + Correct કરી શકે
Checksum	Segments ની Sum + Complement	Burst+Random (TCP/IP)
CRC	Modulo-2 Division (Generator Polynomial)	Burst Errors — સૌથી Powerful
Hamming Code	Redundant Bits ચોક્કસ Position પર	Detect+Correct Single-bit

🚩 CRC Remainder = 0 → No Error | Hamming: $2^r \geq m+r+1$

1 2 Speech Digitization & Coding

Steps: Analog Speech → Sampling → Quantization → Encoding → Digital Output

PCM Standard Parameters (Telephone)

Parameter	Value
Speech BW	300-3400 Hz
Sampling Rate	8000 Hz
Bits/Sample	8
Bit Rate	64 kbps

અન્ય Techniques

Technique	Bit Rate	વિશેષતા
DPCM	<64kbps	Difference Encode
ADPCM	32 kbps	Adaptive Step
Delta Modulation (DM)	એછો, 1 bit/sample	Slope Overload + Granular Noise
Vocoder	~2.4 kbps	Parameter-based (GSM)

1 3 Switching — Space Division & Time Division

Electronic Space Division Switching (ESS)

દરેક Call ને અલગ Physical Path ફાળવાય. Control = SPC (Stored Program Control)
Crosspoints = $n \times m$ (Single Stage) Multi-Stage → Crosspoints ઘટે, Cost ઘટે

Time Division Switching (TDS)

Common Path, Time Slot Share — Digital PCM Systems મળે

TSI (Time Slot Interchange): Speech Memory (Buffer) + Control Memory (Address)

- SWRR (Sequential Write, Random Read)
- RWSR (Random Write, Sequential Read)

T-S-T (Time-Space-Time) ★ — Modern Digital Exchange જે Architecture

SDS vs TDS

Point	SDS	TDS
Path	Dedicated Physical	Shared, Time Slot
Crosspoints	વધુ ($n \times m$)	ઓછા (Memory-based)
વપરિશ્લ	Analog/Digital	ફક્ત Digital (PCM)

1 4 Traffic Engineering

Traffic Intensity (Erlang): $A = C \times T / T_{period}$ ★ 1 Erlang = 1 Circuit અભાવે સમય Busy = 36
CCS = 3600 sec

Busy Hour (BH): સૌથી વધુ Traffic વાળો 1-કલાક Grade of Service (GoS): $GoS = \frac{Lost\ Calls}{Offered\ Calls}$ (Standard = 0.002)

Erlang B: Lost Call System (Queue નહીં) Erlang C: Queuing System (Call Centers)

$Offered\ Traffic = Carried\ Traffic + Lost\ Traffic$

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Telephone Networks

Elements: Subscriber Line, Exchange, Junction, Trunk

Topology

Mesh (Reliable, Costly) | Star (Central-dependent) | **Hierarchical** (Practical — Local→Trunk→Tandem)

India Hierarchy (Traditional 5-Level) ★

LE → TAX → PCC → SCC → CCC (Local Exchange → Trunk Automatic Exchange → Primary → Secondary → Central Class Center)

Exchange ਪ੍ਰਦੇਸ਼

Local Exchange (Direct Serve), Tandem (2 LE ਦਾਇਰਾ), Trunk/TAX (Long Distance), Transit

Numbering Plan: India = Closed Numbering Plan **Signaling:** CCS7/SS7 = Modern Inter-Exchange Signaling Standard

Evolution: PSTN → ISDN → NGN (IP-based)

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Complete Formula Sheet (Last-Minute Revision)

ਕ੍ਰਮ	ਸ਼੍ਰੇਣੀ	Formula	Variables (SI Unit)
1	AM Modulation Index	$m_a = A_m/A_c$	A_m, A_c = Amplitude (V)
2	AM Bandwidth	$BW = 2f_m$	f_m = Modulating Freq (Hz)
3	AM Total Power	$P_t = P_c(1 + m_a^2/2)$	P_c, P_t = Power (W)
4	FM Frequency Deviation	$\Delta f = k_f \times A_m$	k_f = Sensitivity (Hz/V)
5	FM Modulation Index	$\beta = \Delta f/f_m$	$\Delta f, f_m$ in Hz
6	FM Bandwidth (Carson's Rule)	$BW = 2(\Delta f + f_m)$	Hz
7	Image Frequency	$f_{image} = f_{RF} + 2f_{IF}$	f_{RF}, f_{IF} in Hz
8	Image Freq. Rejection Ratio	$IFRR = \sqrt{1 + Q^2 \rho^2}$	Q = Quality Factor
9	Critical Frequency	$f_c = 9\sqrt{N_{max}}$	N_{max} = Max Electron Density
10	Maximum Usable Frequency	$MUF = f_c \times \sec \theta$	θ = Angle of Incidence

ક્રમ	સૂત્ર	Formula	Variables (SI Unit)
11	LOS Distance	$d(\text{km}) = 4.12(\sqrt{h_t} + \sqrt{h_r})$	h_t, h_r = Antenna Height (m)
12	SSB Bandwidth	$BW = f_m$	Hz
13	Nyquist Sampling Rate	$f_s \geq 2f_m$	f_s, f_m in Hz
14	Quantization Levels	$L = 2^n$	n = Bits/Sample
15	SQNR	$SQNR = 6.02n + 1.76 \text{ dB}$	dB
16	PCM Bit Rate	$\text{Bit Rate} = f_s \times n$	bps
17	Entropy	$H = -\sum p_i \log_2(1/p_i)$	bits/symbol, p_i = Probability
18	Coding Efficiency	$\eta = (H/L_{\text{avg}}) \times 100\%$	L_{avg} = Avg Code Length
19	Bit Rate vs Baud Rate	$\text{Bit Rate} = \text{Baud Rate} \times \log_2(L)$	L = Signal Levels
20	Attenuation	$\text{dB} = 10 \log_{10}(P_1/P_2)$	P_1, P_2 = Power (W)
21	Hamming Code Condition	$2^r \geq m+r+1$	m =Data bits, r =Parity bits
22	Traffic Intensity	$A = (C \times T)/T_{\text{period}}$	A in Erlang, T in same unit as T_{period}
23	Grade of Service	$\text{GoS} = \text{Lost Calls} / \text{Offered Calls}$	Dimensionless (Probability)
24	1 Erlang	$1 \text{ E} = 36 \text{ CCS} = 3600 \text{ sec}$	CCS = Centum Call Seconds
25	Crosspoints (Single Stage)	$\text{Crosspoints} = n \times m$	n =Inlets, m =Outlets
26	Offered Traffic	$\text{Offered} = \text{Carried} + \text{Lost}$	Erlang

1 7 Quick Revision & Memory Tricks

✓ Communication System = Source+Transmitter+Channel+Receiver+Destination ✓

AM=Amplitude, FM=Frequency, PM=Phase બંધતીય — Bandwidth ક્રમ: $AM < FM/PM$ ✓ AM Radio: 535–1605 kHz | FM Radio: 88–108 MHz ✓ AM IF=455kHz, FM IF=10.7MHz ✓ FM ની Noise Immunity વધુ — Limiter Circuit ની કાર્યક્ષમતા ✓ Image Frequency = $f_{RF} + 2f_{IF}$; IF ઊંચી → Image Rejection સારું પણ Selectivity ઘટે (Trade-off) ✓ Ground Wave → LF/MF, Sky Wave → HF (Ionosphere), Space Wave → VHF+ (LOS) ✓ SSB=Carrier+1 Side Band Suppress; Power Saving ≈ 83%; Detection=Product Detector+BFO ✓ Manchester Encoding → Ethernet; AMI/HDB3 → PCM Telephone Lines ✓ PAM(Amplitude)

<PWM(Width)<PPM(Position) — Noise Immunity વધતી ક્રમમાં ✓
PCM=Sampling+Quantization+Encoding; Standard Telephone=64kbps ✓ Huffman
Coding=સૌથી Optimal Lossless Compression ✓ FDM=Frequency(Analog),
TDM=Time(Digital), WDM=Optical Fiber, CDM=CDMA ✓ OSI 7 Layers Trick: "**Please Do Not Throw Sausage Pizza Away**" ✓ CRC=Burst Error માટે Best; Hamming
Code=Detect+Correct Single-bit ✓ SDS=Dedicated Path(n×m Crosspoints); TDS=Time
Slot Share(TSI); T-S-T=Modern Exchange ✓ 1 Erlang=1 Circuit પૂરો સમય Busy=36CCS;
GoS Standard=0.002 ✓ India Telephone Hierarchy: **LE→TAX→PCC→SCC→CCC** ✓
India=Closed Numbering Plan; Signaling=CCS7/SS7

1 8 📄 Chapter-Relevant Current Affairs (2025-26)

🔗 GPSC/GSSSB/PSI/Wireless PSI માટે Communication Engineering સાથે સંબંધિત તાજેતરના મુદ્દા

1. **5G in India:** ઓક્ટોબર 2022માં Launch થયેલ 5G હવે India ના 99.9% Districts ને Cover કરે છે (Oct 2025 સુધીમાં). India વિશ્વનું 2જું સૌથી મોટું 5G Market (400+ Million Subscribers, 5.08 લાખ+ BTS Towers).
2. **BSNL 4G/5G Rollout:** BSNL એ ₹1.64 લાખ કરોડનું Government-supported Revival Package (July 2022 Approved) મેળવ્યું છે અને દેશભરમાં Indigenous (TCS/C-DOT આધારિત) 4G Stack Launch કરેલ છે; 5G Rollout 2025-26 દરમિયાન અપેક્ષિત.
3. **National Frequency Allocation Plan (NFAP) 2025:** Department of Telecommunications (DoT) દ્વારા 30 December 2025થી Effective — 6425-7125 MHz Band ને 5G/5G-Advanced/6G માટે Identify કરાયો; Ka, Q, V Bands ને Satellite Broadband માટે Allocate કરાયો.
4. **Bharat 6G Alliance:** ભારત 2030 સુધીમાં 6G Technology માં Global Leadership નું લક્ષ્ય ધરાવે છે.
5. **Satellite Communication (Satcom):** 2026 માં Satcom Connectivity, Fixed Wireless Access (FWA), અને Subsea Cable Landing Station (CLS) Infrastructure (દા.ત. Jio નું Digha, West Bengal ખાતેનું CLS) પર ધ્યાન કેન્દ્રિત.
6. **Telecommunications Act, 2023:** ભારતના Telecom Sector ને Regulate કરતો નવો કાયદો — જૂના Indian Telegraph Act, 1885 નું સ્થાન લે છે.
7. **Digital Infrastructure Investment:** Google (AI Hub, Vizag — Airtel/AdaniConnex સાથે Collaboration), Amazon, Meta જેવી Companies ભારતમાં Data Center Sector માં મોટું Investment કરી રહી છે.

🌟 **Exam Tip:** Current Affairs વિભાગમાં તાજેતરના Spectrum Policy Changes, BSNL/Jio/Airtel ના Rollout Updates, અને Telecommunications Act, 2023 ની જોગવાઈઓ પર ખાસ ધ્યાન આપવું — આ Topics દર વર્ષે અપડેટ થાય છે, તેથી Exam પહેલાં તાજા સમાચાર જરૂર ચકાસવા.

🌟 **Note:** આ Notes Revision Purpose માટે Compress કરેલ છે. વિગતવાર Concept Clarity માટે મૂળ Chapter Explanation ફરી વાંચવી ભલામણપાત્ર છે.

ઝડપી રિવિઝન (Quick Revision) ★

Telephone Network – One Page Revision Notes

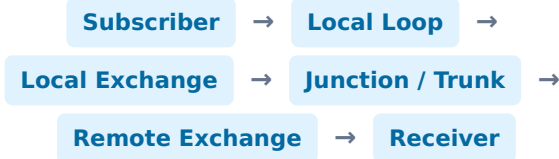
Wireless PSI | GPSC | GSSSB | Class-2/3 Exams

1. મુખ્ય વિષય અને વ્યાખ્યા (KEY CONCEPTS)

વિષય (Topic)	ચાદ રાખવાનો મુખ્ય મુદ્દો (Key Point)
Telephone Network	બે અથવા વધુ ગ્રાહકો વચ્ચે Voice/Data Communication કરાવતું નેટવર્ક
Subscriber Line (Local Loop)	Subscriber ↔ Local Exchange વચ્ચેનું ભૌતિક જોડાણ.
Exchange	Call Switching, Routing, Billing અને Control પ્રક્રિયા કરે છે.
Junction	બે Local Exchange વચ્ચેનું સ્થાનિક જોડાણ.
Trunk	લાંબા અંતર (Long Distance) માટે High Capacity Link.
Mesh Topology	સૌથી વિશ્વસનીય (Reliable), પરંતુ સૌથી ખર્ચાળ (Costly).
Star Topology	બધા Nodes એક Central Exchange સાથે જોડાયેલા હોય છે.
Hierarchical Topology	સૌથી Practical, ઓછો ખર્ચ, મોટા Network માટે શ્રેષ્ઠ.
India Hierarchy Level	LE → TAX → PCC → SCC → CCC
Local Exchange (LE)	Subscriber ને સીધી સેવા (Direct Connection) આપે છે.
Tandem Exchange	બે Local Exchange વચ્ચે Call Forward / Route કરે છે.
Trunk (TAX)	STD / Long Distance Calls પ્રોસેસ કરે છે.
Transit Exchange	અલગ અલગ Exchanges વચ્ચે Routing નું કામ કરે છે
India Numbering Plan	Closed Numbering Plan (નિશ્ચિત ૧૦ અંકો).
SS7 / CCS7	Modern Inter-Exchange Signaling Standard.
PSTN	Traditional Analog Circuit Switched Network.
ISDN	Digital Voice + Data + Fax Integrated Service.
NGN	Modern IP-Based Voice, Video અને Data Network.

2. નેટવર્ક આર્કિટેક્ચર અને ટોપોલોજી (NETWORK DIAGRAMS)

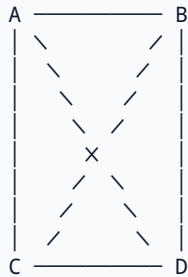
1. Telephone Network Elements Flow



2. Junction vs Trunk Connection

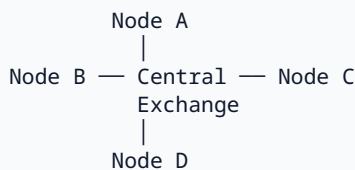
Junction: Local Exchange A ← Junction Link → Local Exchange B
Trunk: Ahmedabad Exch. ← Trunk Line (STD) → Delhi Exch.

3. Mesh Topology (Reliable & Costly)



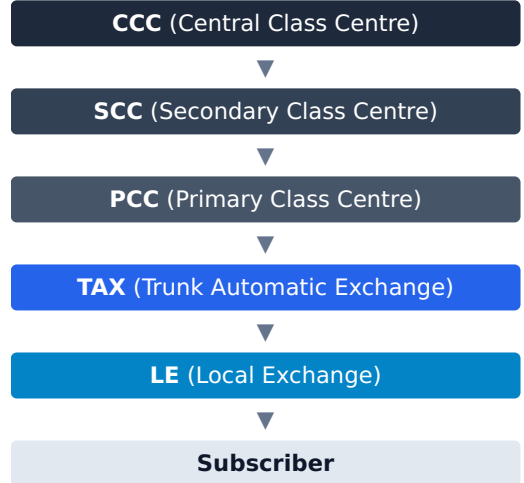
✓ દરેક Node બીજા દરેક Node સાથે સીધો જોડાયેલ છે.

4. Star Topology (Central Switch)



✓ બધા Nodes Central Exchange સાથે જોડાયેલા છે.

5. India Traditional Telephone Hierarchy



6. SS7 Signaling & Evolution

SS7 Signaling:

Exch A — [Voice Channel] — Exch B
Exch A = [SS7 Signaling Link] = Exch B
* Voice અને Signaling અલગ અલગ માર્ગ (Out-of-band) જાય છે.

Network Evolution:

PSTN (Analog) → ISDN (Digital) → NGN (IP Based)

7. India Closed Numbering Plan

9 8 7 6 5 4 3 2 1 0
▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲
Fixed 10 Digits Plan

8. Complete Call Flow Diagram

Caller → Local Loop → LE → TAX → PCC → SCC → CCC → SCC → PCC → TAX → LE → Local Loop → Receiver

💡 શોર્ટકટ ટ્રિક્સ (Memory Tricks for Exam)

📌 **India Hierarchy** યાદ રાખવાની ટ્રિક્સ:

LE → TAX → PCC → SCC → CCC

"લોકલ ટ્રેન પહેલા શહેરના સેન્ટર સિટી પહોંચે" (Local → TAX → Primary → Secondary → Central)

📌 **Network Evolution** યાદ રાખવાની ટ્રિક્સ:

PSTN → ISDN → NGN

Analog → Digital → IP Based

આ રિવિઝન શીટ Wireless PSI, GPSC, GSSSB અને Class-2/3 ની પરીક્ષાઓ માટે શોર્ટકટ રિવિઝન હેતુથી તૈયાર કરવામાં આવી છે.